

DA 210 Project Report

Online Retail Analysis Using PySpark

Introduction

Data (General):

Data refers to a collection of raw facts, figures, or information gathered from different sources. It can exist in various forms such as numbers, text, images, or symbols.

Online Retail Analysis (Specific):

Online Retail Analysis is the process of examining and analyzing sales, customer, and product data from online retail stores to understand customer behavior, improve sales performance, and identify purchasing patterns.

In this project, we used **PySpark** to analyze an **Online Retail dataset**. The purpose of this project is to understand the structure of the dataset, clean the data, and prepare it for further analysis and visualization.

PySpark was selected because it can efficiently process large datasets and provides powerful tools for data manipulation and analysis.



Database Understanding | Lina Ghawanmeh

In this phase, the dataset was loaded and explored to understand its structure and contents.

Imported Libraries

The following **PySpark** libraries were imported:

```
# Import PySpark libraries
from pyspark.sql import SparkSession
from pyspark.sql.functions import *
```

These libraries were used for:

- 1_Creating a Spark session
- 2_Reading CSV files
- 3_Performing data analysis and cleaning operations

* Creating Spark Session:

A Spark session was created using:

```
# Create Spark Session
spark = SparkSession.builder \
    .appName("OnlineRetailAnalysis") \
    .getOrCreate()
```

This step initializes Spark and prepares the environment for data processing.

* Reading the Dataset:

The dataset was loaded from a CSV file:

```
# Read CSV file
df = spark.read.csv(
    "/content/WA_Fn-UseC_-Telco-Customer-Churn.csv",
    header=True,
    inferSchema=True
)
```

Parameters used:

header=True → First row contains column names.

Infer Schema=True → Automatically detects data types.

Displaying Sample Data

The first five rows were displayed using:

```
# Display first 5 rows  
df.show(5)
```

This helped in understanding the dataset structure and checking the data format.

Dataset Schema

The dataset contains **8 columns**:

Column Name	Data Type
Invoice No	String
Stock Code	String
Description	String
Quantity	Integer
Invoice Date	String
Unit Price	Double
Customer ID	Integer
Country	String

After exploring the data:

Number of rows: 541,909

Number of columns: 8

This indicates that the dataset is relatively large and suitable for big data analysis using PySpark.

Data cleaning| Rawaa Alhorani

Data cleaning was performed to improve data quality and prepare it for analysis.

* Checking Missing Values

Missing values were checked using:

```
# Check missing values
from pyspark.sql.functions import col, sum
df.select([
    sum(col(c).isNull().cast("int")).alias(c)
    for c in df.columns
]).show()
```

The results showed missing values in some columns, especially in Customer ID & Description.

* Checking Duplicate Rows

Duplicate rows were checked using :

```
# Check duplicate rows
duplicates = df.count() - df.dropDuplicates().count()

print("Duplicate Rows:", duplicates)
```

Result:

Duplicate Rows = 5268

This means the dataset contained duplicated records that could affect the accuracy of analysis.

Removing duplicate rows:

The dataset was filtered to eliminate any redundant records, ensuring that each row represents a unique customer using:

```
# Remove duplicate rows
df = df.dropDuplicates()
```

To maintain data uniqueness and prevent skewed results caused by duplicated entries.

Customer ID	gender	Senior or Citizen	Partner	Multiple Lines	Dependents	tenure	Phone Service	Internet service
0278-YXWOG	male	0	no	no	5	yes	no	No
9286-BHDQG	Male	0	Yes	yes	72	no	No	DSL
7129-AZJDE	Male	0	Yes	Yes	34	Yes	No	Fiber optic
7024-OHCK	female	1	No	No	2	Yes	yes	Fiber optic

Data Preparation | Layali Alazzam

After completing the data cleaning phase, several preparation steps were performed to make the dataset more suitable for analysis and business interpretation. One of the most important steps was creating a new column called Total Price, which was calculated by multiplying the product quantity by its unit price. This feature provided a clearer understanding of the actual sales value generated from each transaction and allowed deeper analysis of customer purchasing behavior.

In addition, the data was grouped by country in order to calculate the total sales generated from each region. This analysis helped identify the countries with the highest contribution to overall revenue and provided insights into geographical sales performance. Furthermore, the most sold products were identified by calculating the total quantity purchased for each product description. This preparation step helped reveal the products with the highest customer demand and supported further business-oriented analysis.

These preparation techniques enhanced the quality and usability of the dataset and created a strong foundation for the analytical and visualization stages of the project.

Processing Using PySpark Operations| Dana Nawasreh

Several PySpark operations were applied to process and manipulate the dataset efficiently. Different filtering operations were used to extract meaningful subsets of the data, such as identifying products with quantities greater than ten and selecting transactions made within the United Kingdom. These operations allowed the analysis to focus on specific business conditions and customer behaviors.

Moreover, aggregation functions were applied using the `groupBy()` operation to calculate important statistics such as the number of orders per country and the average unit price across different regions. Sorting operations were also implemented to organize products based on unit price values, making it easier to identify highly priced products and analyze pricing patterns.

Finally, the cleaned and processed dataset was exported into a CSV format for future usage and reporting purposes. The use of PySpark significantly improved the efficiency of handling large-scale data and demonstrated the advantages of distributed data processing in real-world analytical tasks.

Analysis & Results| Batol Alghazawi & Rawaa Alhorani

1. Total Sales by Country

This analysis was conducted to identify which countries generated the highest revenue. By calculating the total sales using the `TotalPrice` column and grouping by country, we found the countries contributing most to business revenue. The visualization clearly highlights the top-performing countries and provides insight into geographic sales performance.

2. Most Sold Products

This analysis aimed to identify the most popular products based on quantity sold. By summing product quantities and ranking them, we determined which products had the highest demand. This helps understand customer preferences and inventory priorities.

3. Average Unit Price by Country

The purpose of this analysis was to compare product pricing across countries. By calculating the average unit price for each country, we identified markets associated with higher-priced products. This may reflect purchasing behavior or product variety.

4. Quantity Distribution

A histogram was used to visualize the distribution of purchased quantities. This analysis helps understand customer purchasing behavior and whether customers tend to buy products in small or large amounts.

5. Monthly Sales Trend

This analysis tracked sales performance over time using monthly aggregation. It helps identify seasonal patterns and periods of high business activity by examining changes in total sales month by month.

6. Top Customers by Spending

This analysis identified the highest-spending customers by calculating total purchases per customer. The results help recognize valuable customers and provide insights for customer relationship strategies.

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